

ASHIM DAHAL

Undergraduate Researcher in Computer Vision & Multimodal Learning
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RESEARCH INTERESTS

Computer Vision, Vision-Language Models, 3D Reconstruction, Video Understanding, Multimodal Learning, Neural Rendering (Gaussian Splatting, NeRFs), Efficient Deep Learning, Vision Transformers

EDUCATION

University of Southern Mississippi

Aug 2023 – May 2027 (Expected)

Bachelor of Science in Computer Science; GPA: 3.87/4.0

Hattiesburg, MS, USA

- Keystone Honors Scholar and merit-based full-tuition scholarship
- **Relevant Coursework:** Deep Learning, Computer Vision, Machine Learning, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Software Engineering

SELECTED PUBLICATIONS

Metrics: citations: 30+, h-index: 5

- [C1] A. Dahal, A. Ghimire, S. A. Murad, and N. Rahimi, “POVQA: Preference-optimized video question answering with rationales for data efficiency,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Proceedings Track, 2026.
- [C2] A. Dahal, S. A. Murad, and N. Rahimi, “Embedding shift dissection on CLIP: Effects of augmentations on VLMs’ representation learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Proceedings Track, 2025.
- [J1] A. Dahal, S. A. Murad, and N. Rahimi, “Heuristic comparison of vision transformers against convolutional neural networks for semantic segmentation on remote sensing imagery,” *IEEE Sensors Journal*, vol. 25, no. 10, pp. 17 364–17 373, 2025, [Impact Factor: 4.5].
- [P1] A. Dahal, R. Abdelfattah, and N. Rahimi, *Adaptive anchor policies for efficient 4D Gaussian streaming*, arXiv preprint arXiv:2603.17227, 2026.
- [P2] A. Dahal*, A. Ghimire*, S. A. Murad, and N. Rahimi, *Redemption score: A multi-modal evaluation framework for image captioning via distributional, perceptual, and linguistic signal triangulation*, arXiv preprint arXiv:2505.16180, 2025.

RESEARCH EXPERIENCE

MCTS-VQA: Monte Carlo Tree Search for Video Evidence Collection

Jan 2026 – Present

Undergraduate Researcher

Advisor: Bikramjit Banerjee

- Developing a Monte Carlo Tree Search framework for identifying and collecting question-relevant frame evidence in long-form video question answering.

Adaptive Anchor Policies for Efficient 4D Gaussian Streaming [P1]

April 2025 – Jan 2026

Student Principal Investigator

Advisors: Rabab Abdelfattah and Nick Rahimi

- Developed adaptive anchor-management policies for efficient streaming of dynamic 4D Gaussian representations.
- Investigated policies for anchor insertion, retention, and removal to improve memory efficiency while preserving temporal and rendering quality.
- Independently proposed and led the project through a competitive \$5,500 DCUR Summer Research Grant.

POVQA: Preference-Optimized Video Question Answering [C1]

April 2025 – August 2025

Lead Undergraduate Researcher

Advisor: Nick Rahimi

- Developed a preference-optimized video question-answering framework using rationale-guided training for data-efficient video understanding.

- Reduced input-token requirements by 40% while retaining more than 95% of baseline answer accuracy on long-context video understanding tasks.

Cyber Innovations Lab, University of Southern Mississippi

Aug 2023 – Present

Undergraduate Research Assistant (fully funded); Advisor: Nick Rahimi

Hattiesburg, MS, USA

- **Redemption Score:** Developed a multimodal image-caption evaluation framework combining distributional, perceptual, and linguistic signals for robust image-text alignment assessment. [P2]
- **CLIP Representation Analysis:** Led controlled ablation studies measuring embedding shifts caused by image augmentations in CLIP and related vision-language models. [C2]
- **Remote-Sensing Segmentation:** Compared Vision Transformers and convolutional neural networks for semantic segmentation on satellite imagery and developed a weighted loss function for imbalanced classes. [J1]
- **NASA EPSCoR Project:** Co-designed the research methodology for a \$51,000 NASA EPSCoR-funded project on efficient vision-language models, including objectives, evaluation protocols, and deployment strategies.

HONORS & AWARDS

- \$1,500 Eagle SPUR Research Grant** – Drapeau Center for Undergraduate Research 2026
 - Competitive merit-based research funding for research on intersection of 3DGS and robotics.
- \$500 Best Presentation on Computational Approaches** – Drapeau Center for Undergraduate Research 2026
 - Award winner (First Place) on Undergraduate Symposium for Research and Creative Activity
- \$5,500 Summer Research Grant** – Drapeau Center for Undergraduate Research 2025
 - Competitive merit-based research funding for 4D Gaussian Splatting project (principal student investigator)
- NASA EPSCoR Research Team Member** 2025 – 2026
 - Co-designed \$51,000 funded research project on efficient vision-language models
- Keystone Honors Scholar** – University of Southern Mississippi 2025 – 2027
 - Inducted to the Honors Scholar as a Keystone Scholar for exemplary demonstration of Scholar and Research as an Undergraduate Freshman and Sophomore
- \$500 Hatchery Checkpoint Award** 2024
 - Seed funding for XR application development project targeting dyslexia accessibility
- \$200 Eagles Write Award** – School of Humanities, USM 2024
 - Awarded for best visual analysis essay among all freshman students

LEADERSHIP & SERVICE

- Lead Organizer** 2025 – Present
 - Google Developers Group (GDG) On Campus at University of Southern Mississippi* *Hattiesburg, MS*
 - Delegated USM at Sunnyvale, CA at GOOGLE I/O event in May 2026 (host sponsored)
 - Organize technical workshops, hackathons, and industry speaker events for student developer community
 - Coordinate with Google Developer Relations team to deliver official GDG curriculum and resources
- Research Liaison** 2025 – Present
 - School of Computing Science & Computer Engineering Student Ambassadors* *Hattiesburg, MS*
 - Represent undergraduate research interests to faculty and administration
 - Connect prospective students with research opportunities and faculty mentors
- Head of Artificial Intelligence** 2024 – 2025
 - Google Developer Students Club (GDSC) at University of Southern Mississippi* *Hattiesburg, MS*
 - Led AI/ML workshop series introducing students to deep learning frameworks and practical applications
 - Mentored 20+ students on machine learning projects and research directions

INVITED TALKS & PRESENTATIONS

- “What is an Image?”** – Guest Lecture, Introduction to Computer Science Jan 2025
– University of Southern Mississippi, Hattiesburg, MS
– Introduced foundational concepts of digital images, color spaces, and image processing to 60+ undergraduate students
- “Using AI in Journalism”** – Professional Development Workshop 2023
– Federation of Nepali Journalists, Kaski, Nepal
– Presented practical applications of AI/ML for news verification, content generation, and investigative journalism
- “Future of Artificial Intelligence”** – Career Guidance Seminar 2023
– Fishtail Academy, Pokhara, Nepal
– Discussed emerging trends in AI research and career pathways for high school students

TEACHING & MENTORSHIP EXPERIENCE

- Machine Learning Bootcamp Series** 2022 – 2023
Lead Instructor, Data Research Council for Students *All over Nepal*
– **7-Day Pokhara ML Bootcamp** (2023): Taught 50+ students deep learning fundamentals, CNNs, and PyTorch basics with hands-on projects
– **14-Day Intermediate Python Bootcamp** (2023): Covered advanced Python, object-oriented programming, and ML library usage
– **30-Day Beginner Python Bootcamp** (2022): Introduced 100+ students to programming fundamentals and basic data analysis
– **Chitwan ML Bootcamp** (2023): Delivered intensive week-long training on computer vision and neural networks
- Robotics Instructor** 2021 – 2022
Durbar High School *Kathmandu, Nepal*
– Designed and taught robotics curriculum covering Arduino programming, sensor integration, and mechanical design
– Mentored student teams in regional robotics competitions

SELECTED TECHNICAL PROJECTS & OPEN SOURCE CONTRIBUTIONS

- Torchy** – PyTorch Training Utilities Library 2023
– Developed lightweight PyTorch wrapper library providing clean training utilities (logging, checkpointing, mixed precision, metric tracking) while preserving native `nn.Module` workflows and enabling framework-agnostic rapid prototyping.
– Published to PyPI with comprehensive documentation and example implementations. Adopted by open-source community with 15 stars, 5 forks on GitHub.
– **Technologies:** PyTorch, Python, Weights & Biases integration
– <https://github.com/ashimdahal/torchy>
- ViT vs. CNN Segmentation Benchmark** 2024
– Implemented comprehensive benchmark suite comparing Vision Transformer and CNN architectures for semantic segmentation tasks. Developed novel combined weighted loss function for improved performance on imbalanced remote sensing datasets.
– Structured experiments for reproducibility with detailed ablation studies. Repository directly supports findings published in IEEE Sensors Journal paper (6 stars, 2 forks).
– **Technologies:** PyTorch, Vision Transformers, U-Net, DeepLabV3+
– <https://github.com/ashimdahal/ViT-vs-CNN-Image-Segmentation>
- Jelly** – Low-Resource Language Chatbot 2022
– Created first Romanized Nepali conversational AI using fine-tuned BlenderBot (1.3B parameters) with Flask web interface, targeting accessibility for 30M+ speakers who use Roman script for digital communication.
– Conducted user study analyzing preference for native vs. English language interfaces in mental health contexts. Published findings in Europe PMC.
– **Technologies:** Hugging Face Transformers, Flask, PyTorch, NLP
– <https://github.com/ashimdahal/jelly>

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, SQL, JavaScript, CUDA, Bash

ML/DL Frameworks: PyTorch (DDP, Lightning, FSDP), TensorFlow, JAX, Hugging Face Transformers, Accelerate, scikit-learn, XGBoost

Computer Vision: Vision Transformers (ViT, CLIP, Swin), 3D Gaussian Splatting, Diffusion Models (Stable Diffusion, DALL-E), Neural Radiance Fields (NeRF, Instant-NGP), Video Understanding, Image Segmentation (U-Net, Mask R-CNN), Object Detection (YOLO, Faster R-CNN), Multiview Stereo (MVS)

MLOps & Deployment: Docker, Kubernetes, Git/GitHub, Weights & Biases, MLflow, FastAPI, Flask, Gradio, Streamlit, REST APIs

Cloud & Infrastructure: CUDA Programming, Multi-GPU Training, SLURM, High-Performance Computing (HPC)

Libraries & Tools: NumPy, Pandas, OpenCV, Matplotlib, Plotly, Jupyter, Linux/Unix, Neovim, L^AT_EX, Weights & Biases

Specialized Techniques: Distributed Training (DDP, FSDP), Mixed Precision Training (AMP), Model Optimization, Hyperparameter Tuning, Data Augmentation, Transfer Learning, Few-Shot Learning

PROFESSIONAL AFFILIATIONS

Student Member, IEEE

Member, Google Developer Groups Community

REFERENCES

Available upon request